

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1708

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I __M Kavya 192472370__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: kavya

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

- (a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.
- (b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

- (a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.
- (b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

- (a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.
- (b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?
- (c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

- (a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.
- (b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

(i) Article Citation

Shoaib, L.A., Safii, S.H., Idris, N., Hussin, R., & Sazali, M.A.H. (2024). Utilizing decision tree machine learning model to map dental students' preferred learning styles with suitable instructional strategies. *BMC Medical Education*, 24, 58.

DOI: <https://doi.org/10.1186/s12909-023-05022-5>

Journal: BMC Medical Education (Springer Nature, open access). Note: a title-spelling correction was later issued under DOI 10.1186/s12909-024-05162-2, referring to the same paper.

Domain / application area: Educational technology / dental education

ML problem addressed: Supervised classification — mapping each student's learning-style profile to the most suitable instructional strategy

(ii) PDF / DOI Link

<https://doi.org/10.1186/s12909-023-05022-5>

(Open access — full text and PDF available directly from the publisher page.)

(iii) Written Review

Task 1: Article Summary

(a) Domain and problem addressed

Domain: dental / higher education. Problem: automating the match between a student's learning style and the teaching strategy best suited to them, so that student-centered learning (SCL) can be applied at scale within a dentistry programme.

(b) Summary

Student-centered learning is increasingly expected in higher education, including dentistry, but it has been difficult to apply broadly in dental training. The authors' research gap was practical: instructors have no fast, systematic way to translate a large cohort's individual learning-style profiles into concrete teaching adjustments, and doing this manually for every student does not scale. Their proposed fix was to build a decision tree machine learning model that could take each student's learning-style classification and automatically output a matching set of instructional strategies, effectively creating an instructional-strategy recommender tool for dental educators. To do this, 255 dental students at Universiti Malaya completed a 44-item modified Index of Learning Styles questionnaire, which sorted each student into a learning-style category; this labeled dataset was then used to train a decision tree classifier that automates the learning-style-to-instructional-strategy mapping. The core contribution is turning a subjective, labour-intensive advising task into an instant, reproducible classification pipeline that instructors can apply consistently across an entire cohort.

Task 2: Methodology Analysis

(a) ML technique and dataset

Technique: a decision tree supervised-learning classifier (the specific split-criterion/pruning variant is not detailed in the abstract and should be confirmed from the full-text methods section). Dataset: 255 dental students' responses to the modified Index of Learning Styles (m-ILS), a 44-item questionnaire that classifies respondents along learning-style dimensions. These responses formed the feature set, with each student's learning-style class used as the basis for the target label the tree

was trained to predict (suitable instructional strategy). Source: primary data collected directly from Universiti Malaya dental students — not a public benchmark dataset.

(b) Mapping to CO4 topics; strength and limitation

This is squarely a Decision Trees application, grounded in Inductive Learning principles — generalising a mapping rule from 255 labeled examples to unseen students.

- Strength: Decision trees are inherently interpretable — an educator can read the tree's rules to see why a student was assigned a given strategy, which matters in an advising context where explainability builds instructor trust.
- Limitation: A sample of 255 from a single institution is small and non-diverse (one dental school, one country); decision trees are also prone to overfitting on modest-sized datasets, and the paper does not appear to report cross-validation or an external test cohort, which limits confidence that the rules generalise beyond this cohort.

Task 3: Results & Critical Evaluation

(a) Key results

The IS recommender tool demonstrated perfect precision and recall for overall model accuracy, indicating very high sensitivity and specificity in mapping learning styles to instructional strategies. No competing models were benchmarked, so a comparative table is not applicable — this is a single-model study.

(b) Critical evaluation

- "Perfect" precision/recall is a red flag rather than reassurance in a study like this. It typically signals that the model was evaluated on the same (or very similar) data it was trained on, that the label space is small/simple, or that instructional strategy was assigned to learning style via a fixed rule during data preparation — in which case the tree would trivially "predict" it perfectly.
- Potential biases: single-institution sample, self-reported questionnaire (m-ILS) as the sole feature source, and no mention of a held-out test set or cross-validation.
- Additional experiments that would strengthen the findings: k-fold cross-validation, an external/multi-institution validation cohort, comparison against baseline classifiers (e.g., logistic regression, Naïve Bayes), and a check for label leakage between features and target.

(c) Real-world applicability

The technique could genuinely help academic advisors and curriculum designers triage large cohorts quickly. Scaling challenges include: generalising beyond one institution and one discipline (requires larger, more diverse training data); training faculty to interpret and trust the tool's outputs; and the mild ethical concern of using a single questionnaire snapshot to label students and shape how they are taught, which could inadvertently pigeonhole students if periodic re-assessment is not built into the workflow.

Task 4: Reflection & Learning Outcome

(a) Three key insights

1. Decision trees are attractive in human-facing domains (education, healthcare) specifically because of their interpretability, not just their accuracy — this connects directly to the CO4 Decision Trees topic's emphasis on rule extraction.
2. A "perfect" reported metric is often a methodological warning sign rather than a triumph, reinforcing the CO4 point that evaluation protocol (train/test split, cross-validation) is as important as the choice of algorithm itself.
3. Inductive learning from a small, homogeneous sample constrains how far conclusions can be generalised — a core caution within the Inductive Learning topic regarding hypothesis space and sample representativeness.

(b) Proposed extension

Replicate the study across multiple dental schools (a multi-institution dataset) and add k-fold cross-validation plus a comparison against Naïve Bayes and logistic regression baselines, to test whether the decision tree's apparent perfect performance holds up out-of-sample. This is feasible — it only requires deploying the same 44-item questionnaire at partner institutions and applying standard resampling techniques — and would meaningfully strengthen the claim that the tool is ready for real instructional deployment rather than being an artifact of a small, single-site dataset.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand